

SkinSight AI — Team Project Summary

A simple skin-photo screening app using a trained machine-learning model.

What does our project do?

Users register and log in, upload a skin photo, and receive the model's predicted lesion category with doctor guidance. They can download a PDF report. The app also attempts to send it directly to their registered email through a configured Gmail sender.

Which datasets did we use?

PAD-UFES-20: 2,298 smartphone skin photographs.

ISIC 2019: 24,839 selected dermoscopy images.

Total: 27,137 images covering six categories: actinic keratosis, basal cell carcinoma, melanoma, melanocytic nevus, squamous cell carcinoma, and benign/seborrheic keratosis. ISIC's broader benign category is mapped approximately to the last category.

How was the model trained?

We used **MobileNetV3-Small**, starting with ImageNet-pretrained weights. Images were resized to **160 × 160**. Data was divided into training, validation and test sets, keeping known related patient/lesion images and exact duplicates together.

1. Baseline: Train only the classifier for 5 epochs using both datasets.

2. Fine-tuning: Train all model layers for 5 epochs using both datasets.

3. Smartphone adaptation: Train all layers for 8 epochs using 1,380 PAD training images. The best model was selected at epoch 3.

Fine-tuning used image flips, rotations and color/brightness changes, a batch size of 32, learning rate 0.00003 and the AdamW optimizer. Actual training ran on a CPU.

What results did we get?

Smartphone validation **balanced accuracy improved from 50.4% to 56.9%**. Balanced accuracy averages recall across the six classes. Validation was used to choose the model, so this is not independent test accuracy. A score such as 99.5% shown for one image is a model score, not the project's overall accuracy.

Main features and limits

Features: registration/login, image screening, rejection of obvious icons/blank or tiny images, doctor guidance, PDF downloads and Gmail reports.

Limits: some unrelated photos can still pass the image checks. The model cannot diagnose cancer or determine cancer stage from a photo. Independent clinical testing is still needed; real Gmail sending depends on the sender setup.

Technology: Python, PyTorch, FastAPI, HTML/CSS/JavaScript and SQLite.